

Credit Card Payment System

Team review and demo guide

Frontend	React, Vite, Tailwind CSS
Backend	Django REST Framework and FastAPI
Database	MySQL demo dump included; screenshots use isolated SQLite
Payment mode	Simulated only; no real gateway is connected
Captured	2026-10-07

This review documents the running application using synthetic showcase records. The screenshots do not use the local MySQL records or real payment-card data.

Local showcase accounts: **demo-user** and **demo-admin**. Their passwords are intentionally omitted from this report; they are seeded only in the ignored local SQLite showcase database.

Architecture at a glance

The application separates the customer experience, core account and ledger APIs, and payment simulation. Django is the source of truth for users, card records, and transactions. FastAPI coordinates a simulated payment and calls Django's protected internal endpoints.

Caller	Protocol	Service / responsibility
React + Tailwind	JWT	Django REST: auth, cards, history, admin reporting
React + Tailwind	JWT	FastAPI: validate request and simulate payment
FastAPI	Internal secret	Django internal API: ownership check and transaction state
Django ORM	Database	MySQL in normal configuration; SQLite for screenshots

What happens during a payment

1. The signed-in React client sends the user ID, saved card ID, amount, and JWT to FastAPI. 2. FastAPI verifies the JWT and user identity. 3. Django verifies card ownership and creates a PENDING transaction. 4. FastAPI simulates the outcome and asks Django to finalize it. Django permits only a one-way PENDING-to-final transition.

Security and data handling

Django hashes passwords. Protected APIs require JWT authentication. Card entry validates the number but stores only a masked value and last four digits; CVV is write-only and discarded. FastAPI receives a saved-card ID, never the PAN or CVV. No payment gateway is used.

Review meeting walkthrough

Suggested opening: *This project demonstrates a full-stack payment workflow without moving real money. React handles the customer and admin screens, Django owns authenticated records, and FastAPI simulates payment outcomes.*

Demo sequence

1	Sign in as demo-user and point out the overview metrics and recent activity.
2	Open Cards and show the masked saved card ending in 4242.
3	Submit INR 825.00 for a simulated SUCCESS response.
4	Submit INR 100.13 for the deterministic simulated FAILED response.
5	Open Activity to show both new ledger entries, then review Admin reporting and API docs.

Verification

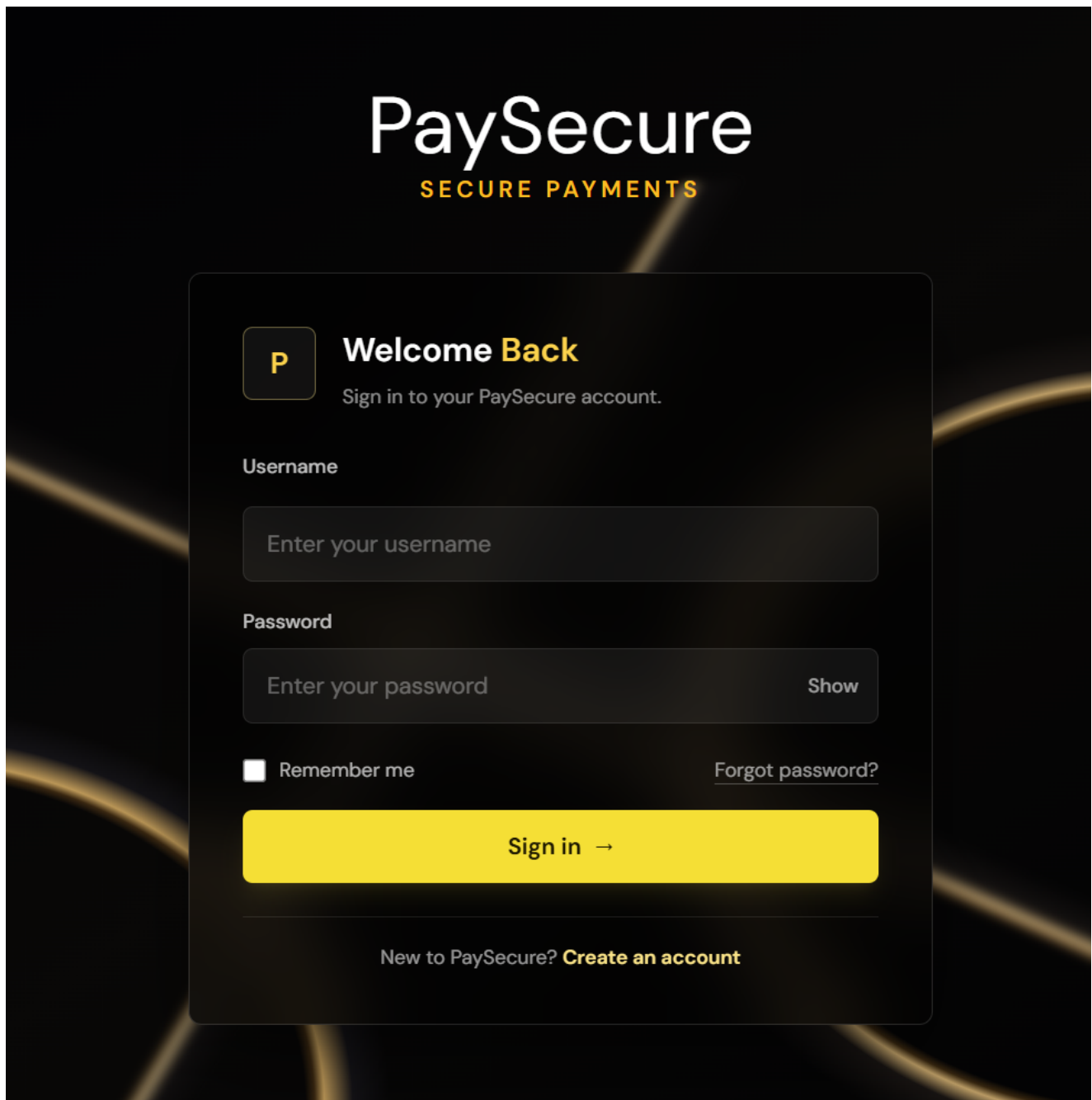
Django	12 tests passed
FastAPI	11 tests passed
Frontend	6 tests passed; production build succeeded
Docker	Not verified in this environment; Docker CLI unavailable

Scope and caveats

Payment outcomes are deterministic demo behavior, not a production processor. The included MySQL dump uses schema-only DDL plus synthetic showcase rows and creates a separate `credit_card_submission_demo` database; existing MySQL rows were not exported or modified. Screenshots use isolated SQLite. Import requires a MySQL administrator because the app account cannot create databases. This guide is not the original assignment PDF.

Complete PNG captures are included in the screenshots folder. The PDF uses a preview crop for layout; open the PNG for the full-page image.

Sign in

The image shows a sign-in interface for 'PaySecure'. At the top, the logo 'PaySecure' is displayed in white, with 'SECURE PAYMENTS' in yellow below it. The background is dark with abstract yellow light streaks. A central dark grey box contains the sign-in form. Inside this box, there's a yellow square with a white 'P' icon, followed by the text 'Welcome Back' in yellow and 'Sign in to your PaySecure account.' in white. Below this are two input fields: 'Username' and 'Password'. The 'Password' field has a 'Show' button to its right. There's a 'Remember me' checkbox and a 'Forgot password?' link. A large yellow 'Sign in →' button is at the bottom of the form. Below the button, a horizontal line separates it from the text 'New to PaySecure? Create an account'.

JWT sign-in screen for the local demo account.

Customer overview

P

PaySecure

Overview

Cards

Payment

Activity

demo-user

YOUR MONEY, AT A GLANCE

Good to see you,
demo-user

Manage cards

Make a payment

A calm, clear view of your payment activity.

Successful volume

₹

₹1,250.00

1 settled payment

Transactions

3

1 pending

Success rate

%

33%

1 failed

Saved cards

1

Card numbers stay masked

Transaction activity

Daily payment attempts over the last seven days

7 days

3

Payment totals, recent activity, and masked saved-card summary.

Dashboard API contract

Credit Card Payment Service 1.0.0 OAS 3.1

/openapi.json

Simulated payment processing service. No real payment gateway is used.

Authorize

default

GET /health Health

GET /version Version

GET /dashboard/summary Dashboard Summary

Parameters

No parameters

Try it out

Responses

Code	Description	Links
200	Successful Response	No links
Media type application/json Controls Accept header: Example Value Schema <pre>{ "available_credit_limit": "42701.00", "current_month_spending": "5149.00", "last_5_transactions": [{ "amount": "-125.50", "card_mask": "*****4242", "created_at": "2026-10-07T10:00:00Z", "currency": "INR", "failure_reason": "", "id": 1, "reference": "TX-20261007-001", "status": "SUCCESS", "updated_at": "2026-10-07T10:00:00Z" }] }</pre>		

FastAPI Swagger documents the dashboard summary route and its response models.

Cards

P

PaySecure

Overview

Cards

Payment

Activity

demo-user

PAYMENT METHODS

Your cards

Keep your payment methods together and ready to use.

1 saved

NEW METHOD

01

Add a card

PAYSECURE

..... 0000

CARDHOLDER NAME12/30CREDIT

Card type

CREDIT

Name on card

e.g. Alex Morgan

Card number

0000 0000 0000 0000

Card entry and saved-card views; only the masked number is persisted.

Simulated payment success

P

PaySecure

Overview

Cards

Payment

Activity

demo-user

SECURE CHECKOUT

Make a payment

A simple, secure way to move money.

Protected payment

PAYMENT DETAILS

01

Where should we send it?

✓

Payment complete

Payment successful (simulated).

Reference PAY-20261005043643-4077FD1E

Pay with

CREDIT . *****4242

Amount

825.00

INR

Payment method

Processing

CREDIT ending 4242

Instant

Pay securely

End-to-end success response from the simulated payment service.

Simulated payment failure

P

PaySecure

Overview

Cards

Payment

Activity

demo-user

SECURE CHECKOUT

Make a payment

A simple, secure way to move money.

Protected payment

PAYMENT DETAILS01

Where should we send it?

! Payment not completed
Payment failed (simulated).
Reference PAY-20261005043702-B2E59174

Pay with

CREDIT . *****4242

Amount

100.13INR

Payment method

CREDIT ending 4242

Processing

Instant

Pay securely

Deterministic failure response from the simulated payment service.

Credit Card Payment System | Local showcase

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Transaction history

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PaySecure

Overview

Cards

Payment

Activity

demo-user

YOUR PAYMENT RECORD

Activity

5 transactions shown

Every payment, clearly accounted for.

Filter activity

Status

From

To

All statuses

mm/dd/yyyy

mm/dd/yyyy

Minimum

Maximum

₹ 0.00

₹ 0.00

REFERENCE	AMOUNT	STATUS	PAYMENT CARD	DATE & TIME
20261005043702-B2E59174	₹100.13	FAILED	*****4242	10/5/2026, 10:07:03 AM
20261005043643-4077FD1E	₹825.00	SUCCESS	*****4242	10/5/2026, 10:06:45 AM
10-PENDING-003	₹780.00	PENDING	*****4242	10/5/2026, 10:04:04 AM
10-FAILED-002	₹340.13	FAILED	*****4242	10/5/2026, 10:04:04 AM
10-SETTLED-001	₹1,250.00	SUCCESS	*****4242	10/5/2026, 10:04:04 AM

Filterable history showing success, failure, and pending states.

Admin reporting

P

PaySecure

Overview

Cards

Payment

Activity

Admin

demo-admin

Admin Dashboard

Transactions

5

All time

Successful

2

All time

Failed

2

All time

Pending

1

All time

Staff-only transaction totals and daily reporting.

Django administration

Django administration

WELCOME, **DEMO-ADMIN**. [VIEW SITE](#) / [CHANGE PASSWORD](#) / [LOG OUT](#) 

Site administration

AUDIT	
Admin logs	+ Add ✎ Change
AUTHENTICATION AND AUTHORIZATION	
Groups	+ Add ✎ Change
Users	+ Add ✎ Change
CARDS	
Cards	+ Add ✎ Change
TOKEN BLACKLIST	
Blacklisted Tokens	+ Add ✎ Change
Outstanding Tokens	✎ Change
TRANSACTIONS	
Transactions	+ Add ✎ Change
Recent actions	

Authenticated Django administration site.

Credit Card Payment System API 1.0.0

OAS 3.0

[/api/schema/](#)

Django API for authentication, cards, transactions and admin reporting.

Authorize



api



GET

/api/admin/export/



GET

/api/admin/summary/



POST

/api/auth/login/



POST

/api/auth/logout/



GET

/api/auth/me/



POST

/api/auth/refresh/



POST

/api/auth/register/



OpenAPI documentation for the Django REST API.

Credit Card Payment Service 1.0.0

OAS 3.1

[/openapi.json](#)

Simulated payment processing service. No real payment gateway is used.

default

GET	/health	Health	▼
GET	/version	Version	▼
POST	/payments/	Make Payment	▼

Schemas

HTTPValidationError >	Expand all	object
PaymentRequest >	Expand all	object
PaymentResponse >	Expand all	object
ValidationError >	Expand all	object

Payment-service Swagger UI, including the typed dashboard summary response.